

Standpoint — The Data Atlas

Every free, public data source behind the globe — what we take from each, and how they connect into one honest world-model.

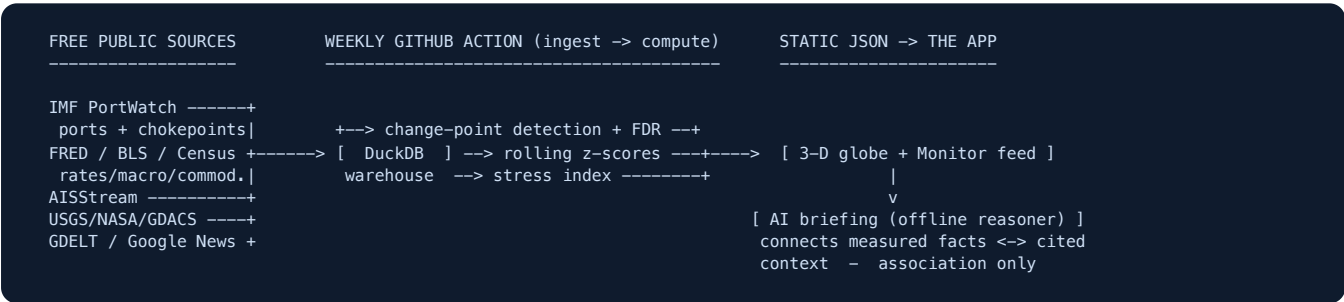
Companion to the [live demo](#) and the [Source Ledger](#) (the same registry, in the app). The full backlog of *candidate* sources lives in [plans/DATA-SOURCES.md](#).

The thesis — in one paragraph

Standpoint takes **only free, public data** — keyless or free-key, never a metered or paid feed — and assembles it into a single 3-D globe of world situational awareness. One domain is the **measured spine** (ocean freight, daily and per-entity, where we run real change-point detection). Everything else is either a **measured signal** (a national index where we compute the anomaly) or **cited context** (someone else's number shown as-is, possibly-related, never a stated cause). An offline AI reads the published result and **connects** the measured facts to the cited context — as association only, never causation, never a forecast. The honesty boundary isn't a promise; it's a compile-time fact (a static import-graph firewall that forbids a context layer from ever writing a measured number).

34 live layers today — 11 spine · 8 signal · 14 context · 1 derived — every one free and cited.

How it connects — the pipeline



No live API at runtime: the weekly Action rebuilds the DuckDB warehouse from source, computes every number in Python, and publishes static JSON. The site is just files — fast, free to host, and impossible to silently change.

The three tiers — who owns the number

Tier	Meaning	Example
SPINE (<i>measured</i>)	The deep, daily, per-entity domain we run real detection on. The flags come from here.	flags , stress from IMF PortWatch
SIGNAL (<i>measured</i>)	A national index where we compute the anomaly (a 12-month rolling z-score), FDR-controlled.	freight_rate (truckload +4.8σ), slack
CONTEXT (<i>cited</i>)	Someone else's number, shown exactly as published — possibly-related, never a stated cause.	quakes , news_geo , wind
DERIVED (<i>AI</i>)	An offline reasoner's prose over the cited store — every claim traces to a layer.	ai_briefing

Promotion context → measured is gated: we only "own" a number when we compute a defensible statistic over raw observed inputs in Python.

The data sources, by category

1 · Ocean freight — the measured spine

The one domain with free, daily, per-entity granularity good enough for real change-point detection. Everything in this block is computed from **IMF PortWatch** (AIS-derived, keyless, refreshed weekly).

Layer(s)	What we take	Cadence	License	Role
IMF PortWatch — ports → snapshot ports timeseries	Daily port-call counts + capacity for ~2,000 ports, by cargo type. We compute STL deseasonalization → 28-day rolling z → PELT change-point → a gated flag.	daily (weekly refresh)	PortWatch terms (free)	spine
IMF PortWatch — chokepoints → chokepoints flags	Daily transit calls for 28 maritime chokepoints by vessel type (container / dry-bulk / tanker / ro-ro) + capacity DWT.	daily (weekly refresh)	PortWatch terms (free)	spine
(derived from the above) stress	Global Ocean Freight Stress Index 0–100 = economic-weighted breadth × worst-chokepoint depth.	weekly	— (computed)	spine
(derived) lanes world events brief	Great-circle lanes, world daily totals, lifecycle events, the templated weekly brief.	weekly	— (computed)	spine
AISStream → ships	A point-in-time sample of live vessel positions near the chokepoints (observed, not all ships).	live sample	AISStream terms (free-key)	context
Panama Canal Authority → gaton	Gatún lake level → percentile → minimum projected Neopanamax draft (ft).	~weekly	ACP terms (free)	signal

2 · Freight cost & macro — measured signals

The cross-domain layer surfaced in the **Signal Board**: trucking, rail, air, inventories, employment. All free **FRED** public-domain series; the number we own is each one's 12-month rolling z-score (FDR-controlled).

Family (layer)	Series we take	Cadence	License	Role
Freight rates freight_rate	Truckload · Rail line-haul · Deep-sea · Air · Warehousing PPI · Freight Transportation Services Index	monthly	public domain (BLS PPI)	signal
Macro / activity macro	Industrial Production · Manufacturing Production · Rail Freight Carloads · Truck Tonnage · Motor Vehicle Production	monthly	public domain (Fed / BTS)	signal
Inventory slack slack	Inventories-to-sales: total business · retail · wholesale · manufacturing · auto	monthly	public domain (Census / BEA)	signal
Transport labor labor	Transportation & warehousing · trade-transport-utilities · truck-transportation · manufacturing employment	monthly	public domain (BLS CES)	signal

3 · Commodities & metals — measured signals

The cargo that moves through the chokepoints, priced. Free **FRED** (IMF Primary Commodity Prices); we own the z-score anomaly, the price stays cited.

Family (layer)	Series we take	Cadence	License	Role
Commodities commodities	Brent crude · EU natural gas · Wheat · Maize (corn) · Soybeans	monthly	public domain (IMF PCPS via FRED)	signal
Metals & bulk energy metals	Aluminum · Iron ore · Nickel · Zinc · Lead · Tin · Copper · Australian coal · Energy Index	monthly	public domain (IMF PCPS via FRED)	signal

4 · Water, canal & marine — context

Physical conditions at the maritime constraints.

Source → layer	What we take	Cadence	License	Role
NOAA CO-OPS → tides	Observed water level at major US ports.	sub-daily	public domain	context
USGS Water Services → streamflow	Observed river stage on freight-relevant rivers.	sub-daily	public domain	context
Open-Meteo (marine) → marine	Model wave height at the chokepoints.	hourly model	CC-BY 4.0	context

5 · Weather & storms — context

Active weather that can physically slow a route — attached to a flag only when contemporaneous, **never** as a stated cause.

Source → layer	What we take	Cadence	License	Role
NHC → weather	Active US tropical-cyclone cones (Atlantic + E/Central Pacific).	per-advisory	public domain	context
GDACS → weather disruptions	Every-other-basin cyclones + global flood/quake/cyclone alerts that hit a monitored port.	event-driven	GDACS terms (free)	context
NOAA GFS → wind	Global 10 m wind field, now → +4 days, baked to small PNGs.	weekly (f000–f096)	public domain	context
NASA GIBS (VIIRS) → satellite	True-color satellite imagery tiles.	daily	public domain	context

6 · Earth hazards & events — context

The "what else is happening on Earth near a place" ring.

Source → layer	What we take	Cadence	License	Role
USGS → quakes	Recent earthquakes (geo + magnitude).	real-time	public domain	context
NASA EONET → eonet	Natural events: fire / volcano / storm / ice.	daily	public domain	context
NOAA SWPC → space_weather	Geomagnetic indices (Kp / solar).	sub-daily	public domain	context

7 · World news — context

Geo-located world news near the chain, and per-flag corroborating headlines.

Source → layer	What we take	Cadence	License	Role
GDELT 2.0 GKG → news_geo	One dot per geo-tagged business/disruption article, by topic.	15-min export	GDELT terms (free)	context
Google News RSS → news	Per-flag curated headlines (linked in the brief's connection lines).	per-refresh	Google News terms	context

8 · Markets — context

Source → layer	What we take	Cadence	License	Role
FRED + Stooq → market	Cited market indicators shown as published, never restated as ours.	daily	public domain (FRED) / Stooq terms	context

9 · Derived — the AI layer

Layer	What it is	Role
ai_briefing	An offline reasoner (Claude Code, zero runtime cost) reads the published store, grounds every claim through <code>verify()</code> , connects measured facts to cited context as association, and is gate-checked fail-closed before it ships.	derived

How the layers connect — the wiring

- 1. **The spine generates the flags.** Only PortWatch has the daily per-entity depth to support change-point detection, so the *alerts* come from maritime. (This is why the straits dominate — by data availability, not by choice.)
- 2. **The signals ride alongside.** The FRED families (freight rate, macro, slack, labor, commodities, metals) are national + monthly — too coarse for per-entity detection, but each yields an FDR-controlled z-score, surfaced in the **Signal Board** next to the flags. "Truckload freight rate +4.8σ" sits beside "Strait of Hormuz -92%."
- 3. **Context corroborates, never creates.** A storm or quake can attach to a flag when it co-occurs in space + time, but it can never *raise* a flag and is never written as a cause. The import-graph firewall makes that a compile error.
- 4. **The AI connects them.** The offline briefing joins a measured disruption to its co-occurring cited news / hazards — "Hormuz -92% co-occurs with 4 cited reports (NPR, CNBC, NYT)" — labelled association, gate-verified.

Gaps & candidates — what to add next, what's missing

The full scouted backlog (93 sources, tagged by tier / auth / cadence / license) is in [plans/DATA-SOURCES.md](#). The highest-value **free**, **unshipped** candidates:

Candidate	Why it matters	Tier it would be	Catch
USACE LPMS (lock performance)	Near-real-time US inland-waterway lock transits + delay minutes — the one <i>non-maritime</i> source with PortWatch-like granularity.	measured (signal/spine)	keyless, but US-only
US Census Intl Trade API	Monthly value + weight by US port & HS code → owned YoY / trend-deviation per port.	measured	free-key, ~5-week lag
OECD CLI / Eurostat / UN Comtrade	National leading indicators, EU PPI + trade, bilateral flows → more macro signals.	measured	monthly/quarterly, coarse
MarineCadastre (NOAA/BOEM)	Annual AIS transit-count raster per US grid cell → owned corridor density delta.	measured	annual cadence
EIA energy	US energy stocks / flows → an energy-logistics signal.	measured	free-key

The structural gap — and why it's honest, not a flaw: the things people most want at *line level* (per-lane truckload rates, container spot rates) only exist as **commercial** feeds — DAT, FreightWaves SONAR, Drewry WCI, Freightos FBX. Pulling them in would break the zero-marginal-cost rule and turn an owned-statistic product into a reseller of someone else's benchmark. So Standpoint stays honest about its shape:

Maritime is the deep measured spine (daily, per-entity). Every other domain is a measured national signal or cited context — broad, but not line-level — because that's the granularity free public data actually offers.

That boundary is the story: not "we have everything," but "we have everything *free*, we're explicit about how deep each layer goes, and we never pretend a context number is a measured one."

Generated from the live layer registry (`backend/freight_radar/registry/`) — the same source of truth the app's [Source Ledger](#) renders. To regenerate the PDF: `python docs/build_atlas_pdf.py` .